

Lightweight Residual Chroma Reconstruction from 4:2:0

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Abstract

Chroma subsampling reduces bandwidth by storing color at a lower spatial resolution than luma, but conventional interpolation blurs color transitions and can misalign them with luminance edges. This paper studies V6, a compact residual convolutional network that reconstructs full-resolution chroma from a synthetic 4:2:0 observation while copying luma exactly. The 593,794-parameter model predicts only a two-channel correction to bilinearly upsampled chroma and is optimized with chroma-only ℓ_1 loss. In a seeded paired pilot on twenty 64×64 COCO crops, V6 obtains 50.5231 dB mean chroma PSNR, versus 47.0957 dB for bilinear interpolation and 46.8985 dB for image-adaptive TabPFN V3. Its paired gains are 3.4274 dB over bilinear (95% bootstrap confidence interval: 2.7926–4.1407 dB) and 3.6246 dB over TabPFN (3.0489–4.2389 dB), with 20/20 wins. An archived 3,703-image evaluation reports a 3.88 dB chroma-PSNR gain, but lacks the original manifest. To close that evidence gap, this revision adds a fully local, hash-addressed protocol covering lossless 4:4:4 sources, a 3×4 siting/filter matrix, actual JPEG/H.264/HEVC/AV1 round trips, classical and learned baselines, controlled ablations, perceptual/color metrics, resource profiling, and difficult-boundary crops. A repository-local 12-image procedural fixture verifies the machinery and produces 1,024 method-image-condition records, including real H.264, HEVC, and AV1 round trips; it is explicitly not scientific evidence. A publication-scale rerun awaits a separately supplied local lossless corpus.

1 Introduction

Many image and video pipelines represent color using a luma channel Y and two chroma channels. In 4:2:0 sampling, each chroma plane has one quarter as many samples as luma. Reconstructing full-resolution color is therefore a structured missing-data problem: edges in Y provide a strong spatial cue, but the missing Cr/Cb values cannot be recovered by interpolation alone.

This work asks whether that structure is better exploited by a small convolutional refiner or by a general tabular foundation model. The repository’s earlier V5 system used a U-Net-like generator and PatchGAN-style discriminator, drawing on common image-to-image translation designs [1, 2]. Its adversarial objective could pro-

duce plausible detail, but its archived pixel-fidelity scores were below bilinear interpolation. Version 6 instead encodes three conservative choices:

- preserve the observed luma exactly;
- begin from a bilinear chroma estimate; and
- learn only the chroma residual under a direct reconstruction loss.

This paper makes three contributions. First, it isolates and evaluates the V6 design against interpolation, V5, and an image-adaptive TabPFN V3 regressor. Second, it introduces an executable publication protocol with immutable source identity, disjoint splits, codec and sampling robustness tests, stronger baselines, ablations, resource measurements, and per-image outputs. Third, it keeps three evidence tiers explicit: a reproducible 20-crop pilot, a large but incompletely archived historical result, and a procedural local audit that validates code paths but is not used to support performance claims.

2 Related Work

Classical reconstruction. Linear and cubic convolution filters remain important low-cost baselines; Keys’ cubic formulation is a canonical image-resampling reference [3]. Edge-aware methods instead use a high-resolution guide: joint bilateral upsampling transfers structure from that guide [4], while guided filtering assumes a local linear relation between guide and output [5]. Because luma is available at full resolution here, both are natural chroma baselines. We therefore compare nearest, bilinear, bicubic, Lanczos, guided-filter, and joint-bilateral reconstruction under identical low-resolution samples.

Learned restoration and chroma guidance. SR-CNN established a compact three-layer learned super-resolution baseline [6]; EDSR later showed the value of deep residual blocks without unnecessary normalization [7]. More recent restoration families include windowed transformers such as SwinIR [8] and nonlinear-activation-free blocks such as NAFNet [9]. Chroma upsampling is not merely generic RGB super-resolution: Li et al. explicitly used luma to guide a learned chroma upsampler for intra coding [10]. The revised benchmark consequently

includes chroma-adapted SRCNN and shallow NAF-style trainable controls in addition to V5 and the EDSR-like V6 residual body.

Compression-aware restoration. JPEG commonly couples transform quantization with chroma subsampling [11]; H.264/AVC, HEVC, and AV1 introduce distinct prediction, transform, and filtering behavior [12, 13, 14]. Approximating a codec can miss consequential implementation details, as work on differentiable JPEG demonstrates [15]. Learned decoders can also target either distortion or perceptual plausibility [16]. This motivates testing actual locally encoded bitstreams at several operating points rather than inferring codec performance from a single synthetic operator.

Synthesis and general tabular priors. Encoder-decoder architectures such as U-Net retain multi-scale detail [1]; conditional adversarial models can encourage locally realistic outputs [2], but realism and exact reconstruction are different objectives. TabPFN is a prior-data fitted network for small-to-medium tabular prediction [17]. Our pilot represents pixels as rows of spatial and luma-derived features. This is flexible and needs no project-specific offline training, but does not encode translation-equivariant neighborhoods as directly as convolution.

3 Residual Chroma Refiner

3.1 Synthetic 4:2:0 observation model

Let the full-resolution image in normalized YCrCb coordinates be $I = [Y, C]$, where $C = [C_r, C_b]$. The degradation operator keeps Y unchanged, area-downsamples each chroma channel by two in both dimensions, and restores it to the original size with bilinear interpolation:

$$\tilde{C} = U_{\text{bilinear}}(D_{\text{area}}(C)), \quad X = [Y, \tilde{C}]. \quad (1)$$

Thus all compared methods receive chroma derived from the same low-resolution samples. This is an idealized simulation rather than a complete codec model.

3.2 Architecture and objective

The refiner maps the three-channel tensor X to 64 features with a 3×3 convolution. Following residual optimization principles [18], eight blocks each apply convolution, ReLU, and convolution before an identity addition. A final 3×3 convolution predicts a two-channel residual $R_\theta(X)$. The reconstruction is

$$\hat{I} = [Y, \tilde{C} + R_\theta(X)]. \quad (2)$$

All convolutions have unit stride and preserve resolution. The network contains 593,794 trainable parameters and has a maximum theoretical receptive field of 37×37 pixels. Figure 1 summarizes the dataflow.

Training minimizes mean absolute error only on chroma:

$$\mathcal{L}_{\text{chroma}}(\theta) = \frac{1}{2HW} \left\| \tilde{C} + R_\theta(X) - C \right\|_1. \quad (3)$$

The reference implementation uses random 256×256 crops, a 97/3 train/validation split, AdamW with learning rate 10^{-4} , batch size 16, and 30 epochs. Training used one NVIDIA RTX 6000 Ada GPU in a workstation equipped with two RTX 6000 Ada GPUs. The tested checkpoint is `res_epoch_030.pth`.

4 Experimental Protocol

4.1 Existing paired COCO pilot

The primary completed comparison uses twenty 64×64 crops from COCO 2017 validation images [19]. Sorted image paths are shuffled with seed 2026; one uniformly located crop is then drawn from each selected image. The same crop, color conversion, and simulated 4:2:0 observation are used for every method. The compared methods are bilinear interpolation, bicubic interpolation, the epoch-10 V5 checkpoint, the epoch-30 V6 checkpoint, and TabPFN V3.

For TabPFN, every full-resolution pixel is a test row with 16 features: normalized coordinates; polynomial and sinusoidal coordinate terms; luma; two Gaussian smoothings; a luma high-pass residual; horizontal and vertical luma derivatives; and gradient magnitude. Area-downsampled features and observed chroma form 1,024 training rows. Two hosted regressors (identifier `v3_default`) independently predict C_r and C_b for 4,096 test rows. Consequently, TabPFN is fit anew for each crop, whereas V6 performs a single feed-forward pass with fixed weights.

4.2 Publication-scale data contract

The expanded runner accepts only decoded RGB/RGBA images in lossless PNG, TIFF, BMP, or PPM containers. Thus every source is a full-resolution 4:4:4 raster, not a JPEG decoded before the reference is defined. A JSONL manifest records the relative path, split, SHA-256 digest, dimensions, mode, format, and source group of every image. Hashes are recomputed before training and evaluation; an identical digest in different splits aborts manifest creation. Train, validation, and test directories are physically separate, while scene-family and near-duplicate exclusion remains an explicit curation audit. Lossless restoration corpora such as DIV2K provide a precedent for this source design [20]. The full manifest must be committed before its numbers enter the paper.

4.3 Sampling, codecs, and baselines

Synthetic evaluation crosses three chroma locations—centered, horizontally left-sited, and fully cosited—with

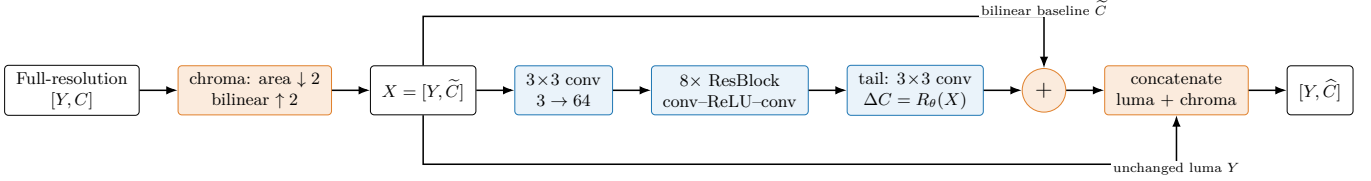


Figure 1: Version 6 predicts a chroma correction ΔC and adds it to the bilinear baseline $\tilde{C}$. Luma bypasses the CNN unchanged. Identity skips are also present inside every residual block.

Table 1: Preregistered V6 ablation matrix.

Factor	Base	Alternatives
Luma guidance	on	off
Global residual	correction	direct chroma
Block skips	on	off
Depth (blocks)	8	4, 12
Width (features)	64	32, 96
Loss	ℓ_1	MSE, Charbonnier, ℓ_1 +edge
Architecture	V6-style	SRCNN, NAF-style

box, triangle, Gaussian, and Lanczos-3 prefilters, yielding twelve conditions. The compared classical reconstructors are nearest, bilinear, bicubic, Lanczos-3, guided filtering, and joint bilateral upsampling. Learned methods include the committed V5 and V6 checkpoints, a retrained V6-style control, a chroma-adapted SRCNN, and a shallow NAF-style restoration model. All methods in a condition receive exactly the same low-resolution chroma.

Actual codec paths complement the synthetic grid. Pillow encodes baseline JPEG with explicit 4:2:0 sampling at qualities 30, 50, 70, and 90. When the requested local FFmpeg encoders exist, H.264 and HEVC are evaluated at CRF {18, 23, 28, 35} and AV1 at {20, 30, 40, 50}, always forcing yuv420p. Missing executables or encoders produce structured skips. These are single-frame round trips, so they test spatial codec artifacts but not temporal stability.

4.4 Controlled ablations

Every ablation uses the same manifest, crop schedule, degradation draw, AdamW settings, seed, and validation checkpoint rule. Table 1 specifies one-factor comparisons; the SRCNN and NAF-style entries are separate architecture controls and are not used for causal attribution.

4.5 Metrics, statistics, and resource cost

PSNR is computed on values normalized to $[0, 1]$ and capped at 80 dB. SSIM uses an 11×11 Gaussian window [21]. The expanded report contains RGB and chroma PSNR/SSIM, chroma MAE, strongest-quartile edge MAE, gradient MAE, and mean and 95th-percentile CIEDE2000 [22]. It also records mean/median/p95 synchronized runtime, trainable parameters and bytes, convolutional FLOPs (a multiply and add count as two), Python allocat-

ion peak, process-RSS delta, and accelerator allocation peak. Codec round-trip time and encoded bytes are labeled separately from network inference.

Comparisons are paired by image and condition. Mean directional differences include win/tie rates and percentile bootstrap intervals [23]: 2,000 resamples for the existing pilot and 10,000 for the full protocol. Every observation is written to `per_image.jsonl`; aggregates are derived from that file. Hard examples are ranked by baseline (bilinear or codec-native) chroma-edge error, and crops center on the strongest reference chroma gradient.

4.6 Evidence tiers and local audit

The paired pilot is reproducible but small. The separate 3,703-image record has scale but no original manifest or per-image output, so it remains historical and is not pooled with the pilot. The new publication protocol is implemented but has not been run on a natural-image corpus because none is present locally. Instead, a repository-local procedural audit uses twelve deterministic lossless 96×96 patterns split 6/2/4. It verifies the manifest, training, metrics, profiling, JPEG, paired aggregation, and qualitative paths. The audit writes 1,024 method-image-condition observations across all twelve synthetic conditions, four JPEG qualities, and four operating points each for H.264, HEVC, and AV1, plus 32 boundary sheets. All 48 requested video-codec image-settings completed through a locally installed FFmpeg, with zero skips. These generated patterns validate software only; no score from them is a research result.

5 Results

5.1 Paired pilot

Table 2 shows that V6 has the best mean on every reported chroma measure. It exceeds the strongest interpolation baseline, bicubic, by 2.3658 dB and exceeds TabPFN by 3.6246 dB in chroma PSNR. The MAE results agree with PSNR: V6 reduces mean chroma error by 33.8% relative to bilinear and 27.8% relative to TabPFN. Its edge MAE is 39.5% lower than bilinear and 28.8% lower than TabPFN, suggesting that the gain is not confined to flat regions.

The paired analysis in Table 3 is stronger than a comparison of means alone. V6 beats bilinear and TabPFN

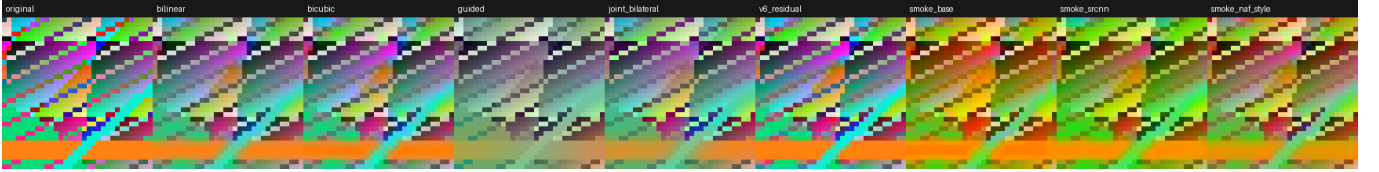


Figure 2: Automatically selected difficult-boundary crop from the procedural fixture audit. Panels compare the lossless reference with classical, V6, and trainable smoke-model reconstructions under centered box-filtered 4:2:0. Nearest-neighbor display enlargement exposes chroma transitions. The synthetic pattern and one-epoch smoke checkpoints make this a pipeline validation image, not qualitative evidence of natural-image performance.

Table 2: Mean results on twenty paired COCO crops. Best values are bold. Error metrics are lower-is-better.

Method	PSNR	SSIM	MAE	Edge MAE	Grad. MAE
Bilinear	47.0957	0.951135	0.005202	0.008581	0.002587
Bicubic	48.1573	0.959445	0.004745	0.007726	0.002292
V5	42.9576	0.959965	0.006497	0.008563	0.002380
TabPFN V3	46.8985	0.961636	0.004773	0.007287	0.002342
V6	50.5231	0.978180	0.003444	0.005190	0.001569

Table 3: Paired chroma-PSNR advantage of V6. Positive values favor V6; CIs use 2,000 bootstrap resamples.

Reference	Gain (dB)	95% CI (dB)	Wins
Bilinear	+3.4274	[2.7926, 4.1407]	20/20
TabPFN V3	+3.6246	[3.0489, 4.2389]	20/20

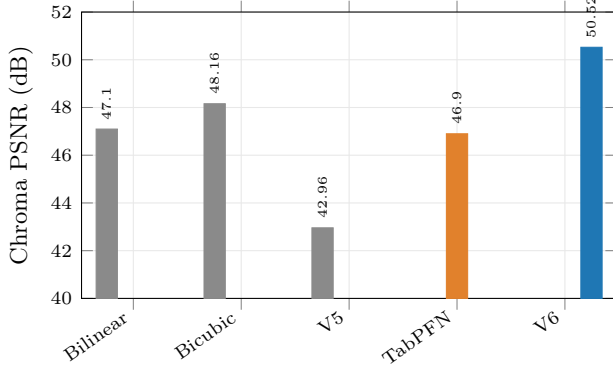


Figure 3: Mean chroma PSNR in the paired COCO pilot. The truncated axis is labeled explicitly to emphasize method differences.

on all twenty crops in chroma PSNR. The confidence intervals exclude zero by wide margins. Against TabPFN, V6 also wins all crops on chroma SSIM, MAE, edge MAE, and gradient MAE.

TabPFN does show a different strength: relative to bilinear, it reduces average edge MAE by 0.001293, with a 95% interval of 0.000025–0.002893. However, it wins only 40% of individual crops on that metric, and its mean chroma PSNR is 0.1972 dB lower than bilinear with an interval spanning zero. Thus the tabular formulation can align some chroma transitions with luma cues, but its benefit is image-dependent in this pilot.

5.2 Archived 3,703-image evaluation

The larger historical record in Table 4 is consistent with the pilot. V6 improves mean RGB PSNR by 3.62 dB and chroma PSNR by 3.88 dB over bilinear interpolation. Minimum PSNR also improves, although the 80 dB cap

Table 4: Archived evaluation on 3,703 validation images. These aggregate values lack a committed manifest and per-image report.

Metric	Method	Mean	Min.	Max.
RGB PSNR	Bilinear	40.42	24.19	80.00
	V6	44.04	25.50	80.00
Chroma PSNR	Bilinear	42.88	26.83	80.00
	V6	46.76	28.29	80.00

is reached by both methods on some images.

The archived video experiment reports PSNR increasing from 57.54 to 58.55 dB and SSIM from 0.9985 to 0.9988. Without temporal state or a committed source video, this shows only frame-level compatibility.

5.3 Why the conservative design helps

Three observations explain the result. First, copying Y makes RGB fidelity monotonic with chroma quality for a fixed conversion; the model cannot damage already observed luminance. Second, predicting a residual makes bilinear interpolation an easily retained default in smooth or uncertain regions. Third, the convolutional body encodes locality and translation equivariance directly. TabPFN receives engineered spatial features, but it must infer neighborhood behavior from an unordered table and fit two regressors for every image. The results support the task-specific inductive bias of V6 over a more general tabular prior for this reconstruction setting.

The V5 comparison reinforces the loss-design argument. V5 attains slightly higher chroma SSIM than bilinear in the archived record (0.9536 versus 0.9530), yet loses 1.90 dB chroma PSNR and 3.31 dB RGB PSNR. Adversarial synthesis can favor plausible structure without preserving the exact target samples. For chroma reconstruction

aimed at objective fidelity, direct residual regression is better aligned with the evaluation criterion.

6 Limitations and Future Work

The implementation closes the earlier protocol omissions, but not the evidence gap by itself. Twenty crops remain too few to characterize COCO, and their 64×64 size favors manageable TabPFN tables rather than full-resolution scaling. The historical large result is still unauditable. Most importantly, the new large natural-image benchmark has not been executed: the repository contains neither a licensable lossless corpus nor the missing historical sources. Although all three local codec encoders complete on the procedural fixture, substituting generated patterns for a natural corpus would create a misleading result, so their scores remain excluded from the claims.

SHA-256 prevents exact split duplication but cannot detect altered near-duplicates or frames from the same scene; those require corpus-level grouping and perceptual-duplicate review. The codec experiment processes independent frames and decoded RGB, not native decoder chroma planes, so it cannot establish temporal consistency or bit-exact chroma behavior. Runtime and memory are hardware- and backend-dependent, and the reported convolutional FLOP count excludes non-convolutional operations. Finally, the NAF-style control is a shallow same-resolution adaptation, not a reproduction of the full NAFNet encoder–decoder.

Completion now requires execution rather than further metric design: supply and curate the local lossless corpus, commit its manifest, train the preregistered matrix, enable the local codec encoders, run the full configuration, and commit the per-image report and boundary crops before replacing any table. A later video study should add multi-frame sequences and a temporal-consistency metric.

7 Conclusion

A 593,794-parameter residual CNN provides a strong, simple solution to synthetic 4:2:0 chroma reconstruction. On the paired COCO pilot, V6 improves chroma PSNR by 3.43 dB over bilinear interpolation and 3.62 dB over image-adaptive TabPFN, winning every crop in both comparisons. The archived 3,703-image result reports a similar 3.88 dB advantage over bilinear. These gains arise from a useful alignment of architecture and objective: preserve luma, retain bilinear as the base case, exploit local spatial structure, and learn only the missing color correction. The revised repository makes the essential next test audit-ready—including source identity, codec diversity, strong baselines, ablations, perceptual color error, cost, and qualitative boundaries—while leaving publication-scale cells empty until real local data are available.

Reproducibility Statement

The implementation and committed checkpoints are available in the Neural Chroma Reconstruction repository. The paired pilot is documented at revision `f4c8563` and is generated by `scripts/benchmark_coco_tabpfn.py` with 20 images, crop size 64, seed 2026, the committed V5 and V6 checkpoints, hosted model identifier `v3_default`, and 2,000 bootstrap samples. Hosted TabPFN execution requires the corresponding client credentials. The historical values are transcribed from the project README and are explicitly labeled non-reproducible in this paper. The expanded local protocol is defined by `research/configs/benchmark_full.json` and `research/configs/ablations.json`. Its procedural audit is reproducible from `research/manifests/fixture.jsonl`; outputs are stored under `research/reports/smoke`, including all 1,024 JSONL records and the crop shown in Figure 2. No network service is used by that audit.

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